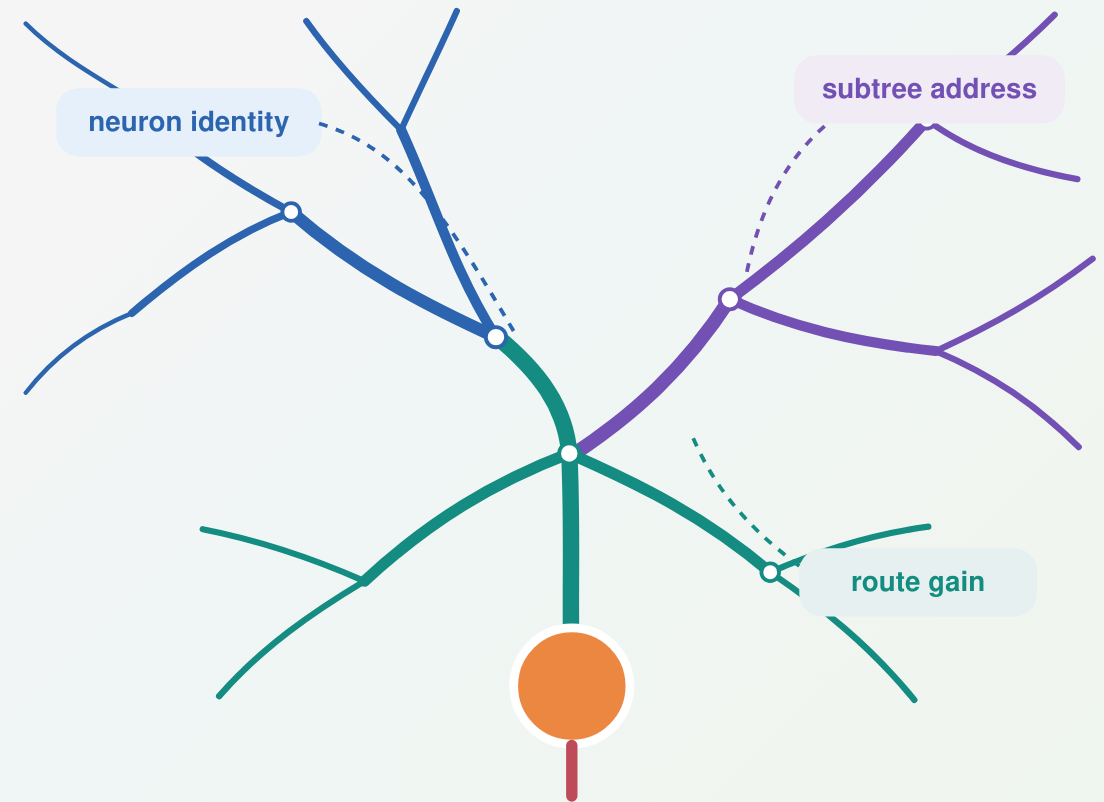


When dendritic structure helps local credit assignment

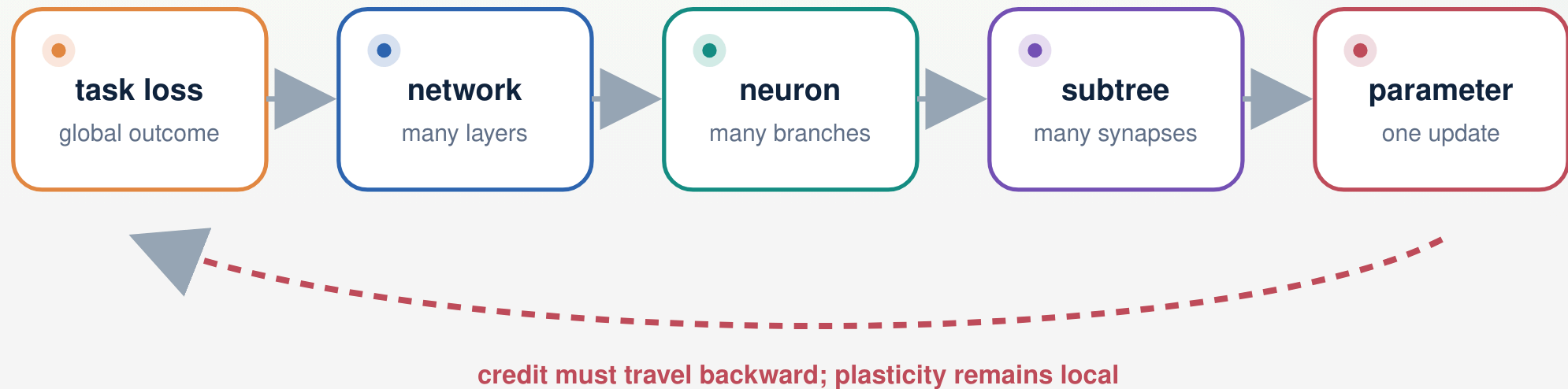
From neuron-level errors to subtree addresses, route gain, and an alignment boundary

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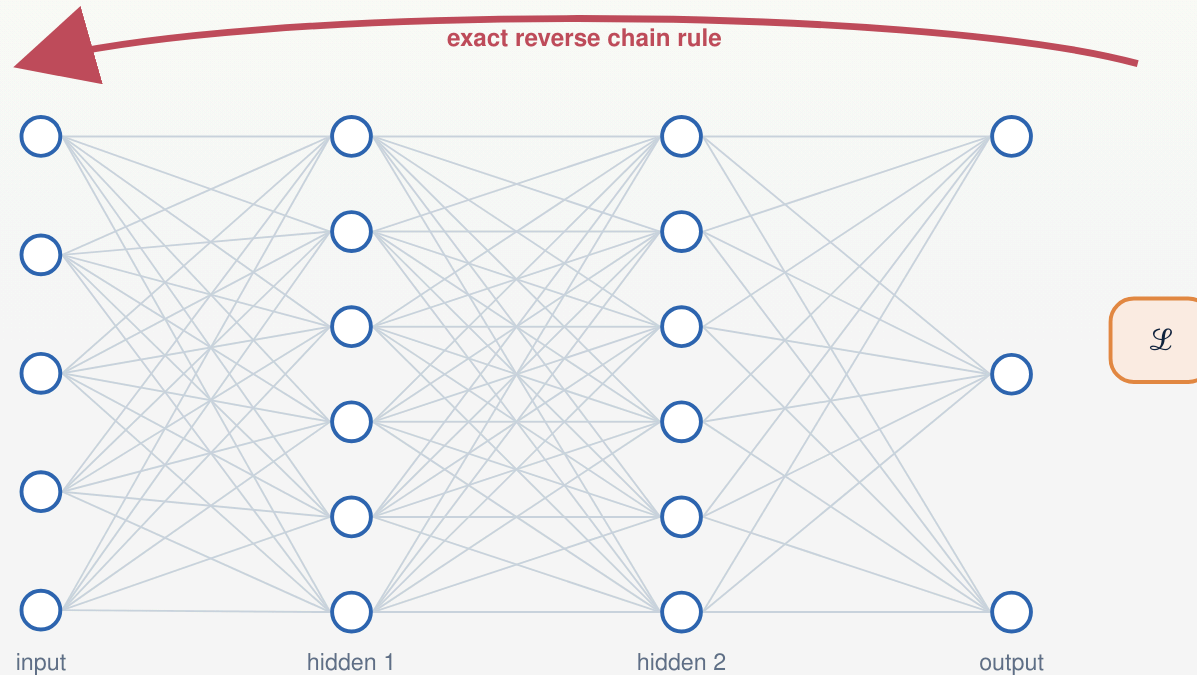
Credit assignment is a routing problem



$$\text{parameter credit} = \partial \mathcal{L} / \partial \theta_i$$

→ A circuit computes one global outcome, but learning must reach the local parameters that caused it.

Backpropagation provides exact parameter credit



$$\Delta W_{ji}^{\ell} = -\eta \cdot h_i^{\ell-1} \cdot \delta_j^{\ell}$$

local factor

Presynaptic activity and the derivative at the receiving unit.

nonlocal factor

A parameter-specific error assembled by the reverse chain rule.

We use backpropagation to define the target information—not as a claim about literal neural implementation.

➔ Backpropagation answers three questions for every parameter: destination, sign, and magnitude.

A local rule preserves eligibility and approximates the error

$$\Delta w_i = -\eta$$

 e_i

synapse-local eligibility

 $\times$ **δ_u**

communicated learning signal

local

e_i can use presynaptic activity, voltage, driving force, and gates.

task-dependent

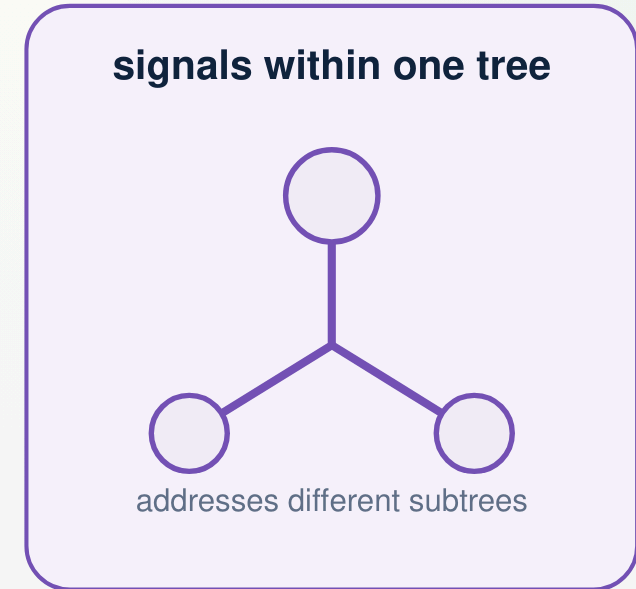
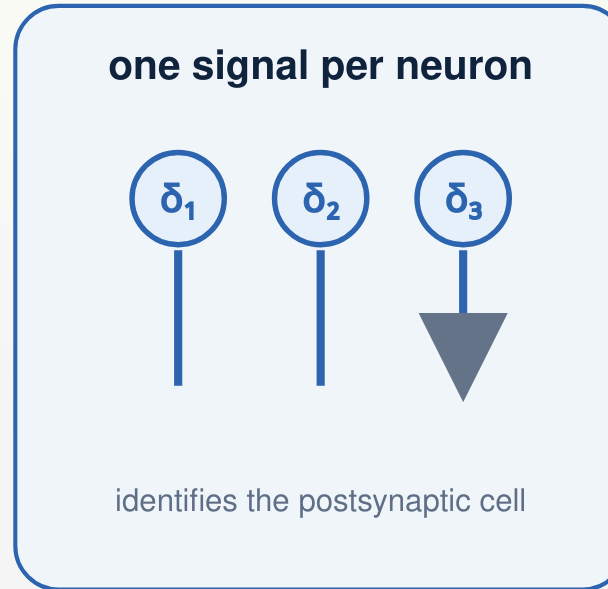
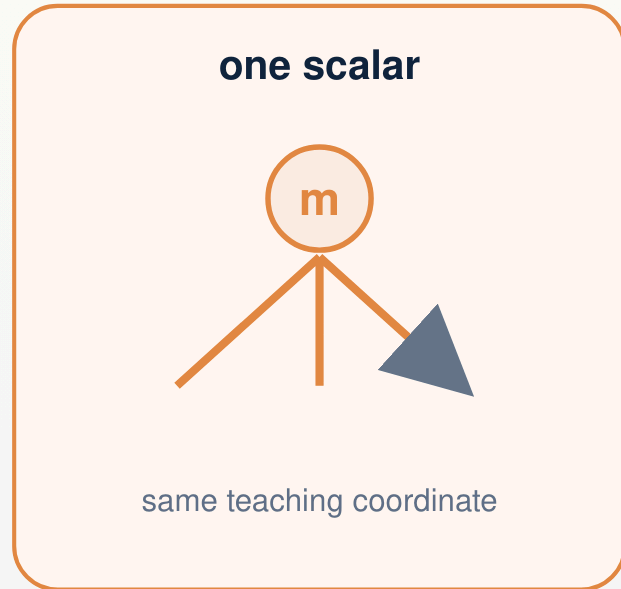
δ_u may depend on a global objective even when the synaptic update is local.

information-limited

The feedback pathway determines which distinctions the rule can express.

→ **Locality specifies where an update is computed; feedback bandwidth specifies what the update can know.**

The first bottleneck is which neuron should learn



Important: a shared scalar does not make neurons identical. Each synapse still has its own eligibility and initialization; the scalar only removes task-specific feedback coordinates.

→ **Neuron identity and within-neuron address are different credit-assignment problems.**

A dendritic arbor adds state, addresses, and route gain



1 · local state

Voltages and conductances change the eligibility computed at each synapse.

ML analogy: feature-dependent local Jacobian

2 · subtree address

A small number of signals can target nested groups of synapses.

ML analogy: structured low-rank routing

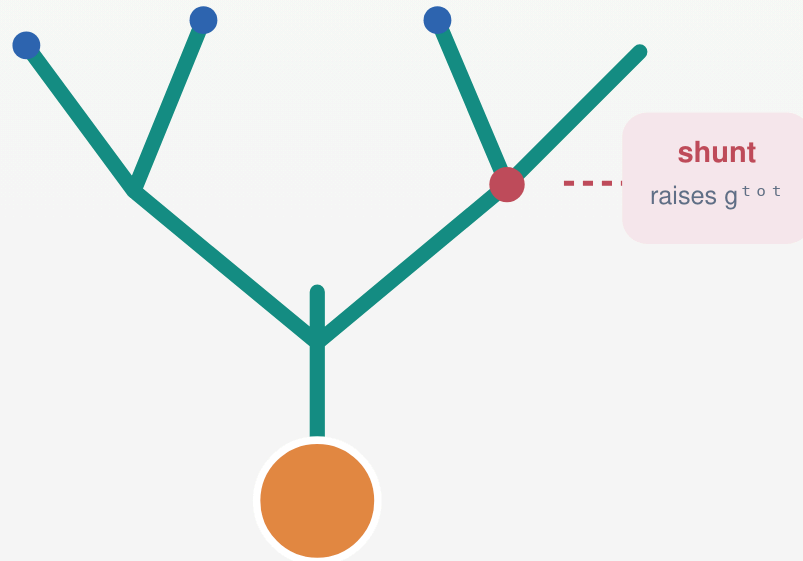
3 · route gain

Conductance can scale how strongly error propagates along one path.

ML analogy: state-dependent feedback preconditioner

→ The question is not whether dendrites add parameters—it is whether their spatial structure matches the credit the task requires.

Conductance makes dendritic voltage a normalized quotient



$$V_n = \frac{g^L E^L + \sum_i g_i x_i E_i + \sum_c g_{c \rightarrow n}^{den} a_c}{g^L + \sum_i g_i x_i + \sum_c g_{c \rightarrow n}^{den}}$$

$R_n^{tot} = 1 / g_n^{tot}$ is the local input resistance.

additive current

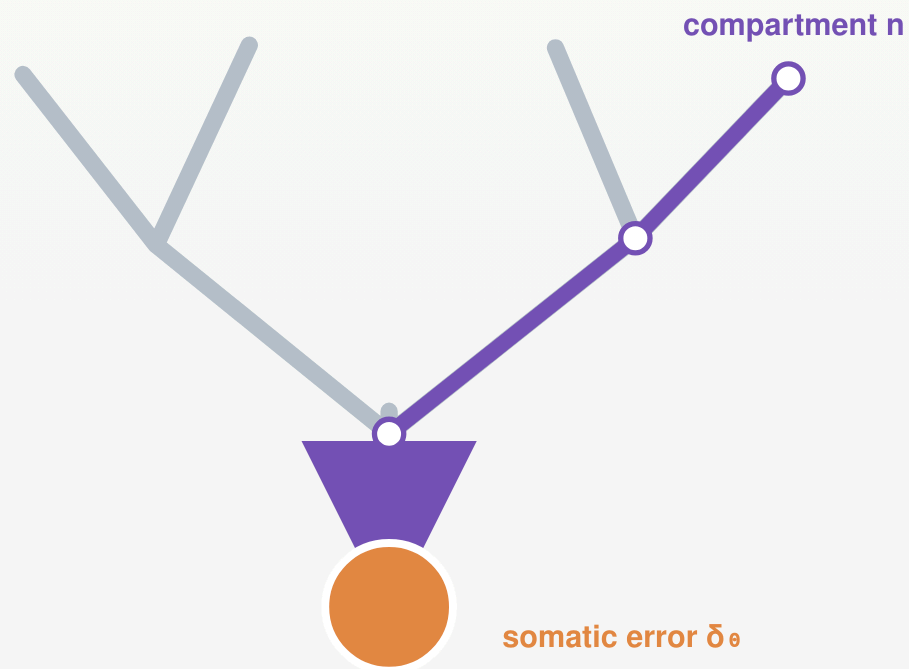
Can match the local voltage change without changing g^{tot} .

shunting conductance

Changes both voltage and the denominator that controls sensitivity.

→ A shunt can change the gain of other synapses even when its own net current is small.

Eligibility stays local; the error becomes a field over the arbor



$$\frac{\partial \mathcal{L}}{\partial g_i} = \underbrace{x_i R_n^{\text{tot}} (E_i - V_n)}_{\text{local dendritic eligibility } e_i} \times \underbrace{\frac{\partial \mathcal{L}}{\partial V_n}}_{\text{compartment error}}$$

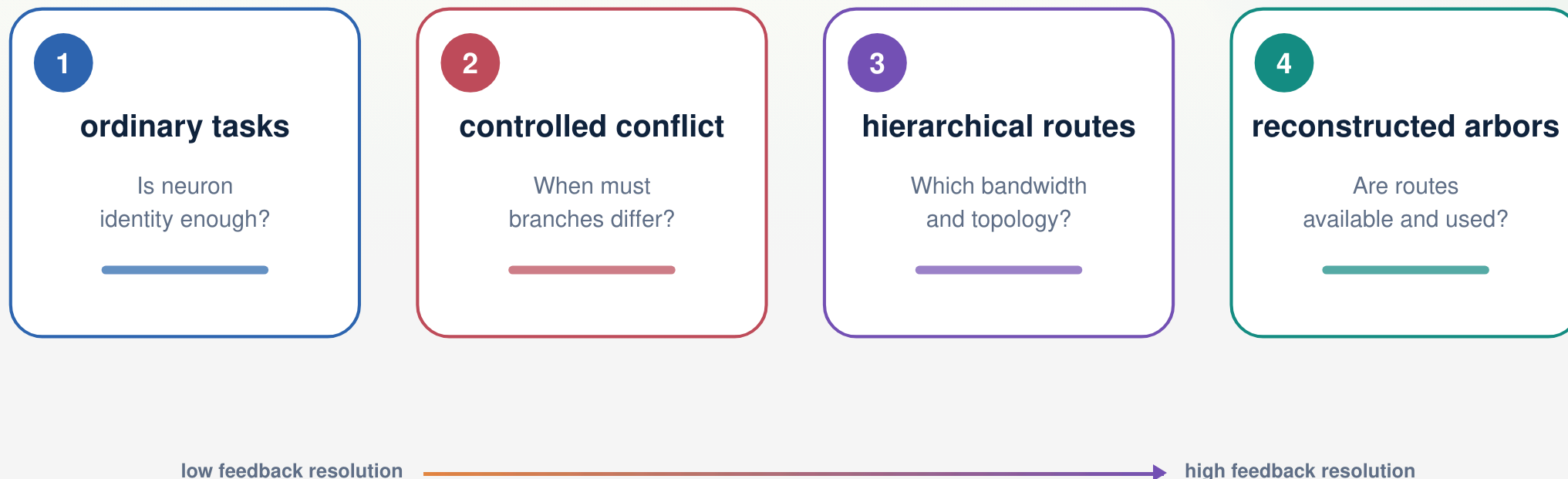
$$\frac{\partial \mathcal{L}}{\partial V_n} = \delta_{0,u} \cdot \alpha_n$$

$$\alpha_n = \prod_{(i \rightarrow k) \text{ on path}(n \rightarrow 0)} f'_i(V_i) R_k^{\text{tot}} g_{i \rightarrow k}^{\text{den}}$$

In a general compartmental model, the same transported field is obtained from the steady-state adjoint $J_V^T q = \nabla_V \mathcal{L}$.

→ Dendrites do not create the circuit-level error; they can distribute it through spatially structured gains and addresses.

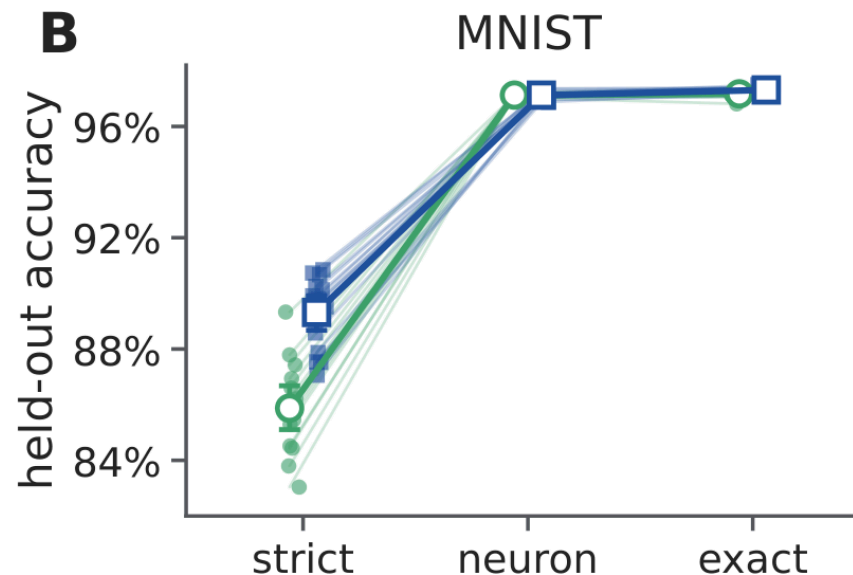
We test the need for progressively richer credit



→ Every positive result is paired with a matched point implementation to separate the information resource from dendritic material.

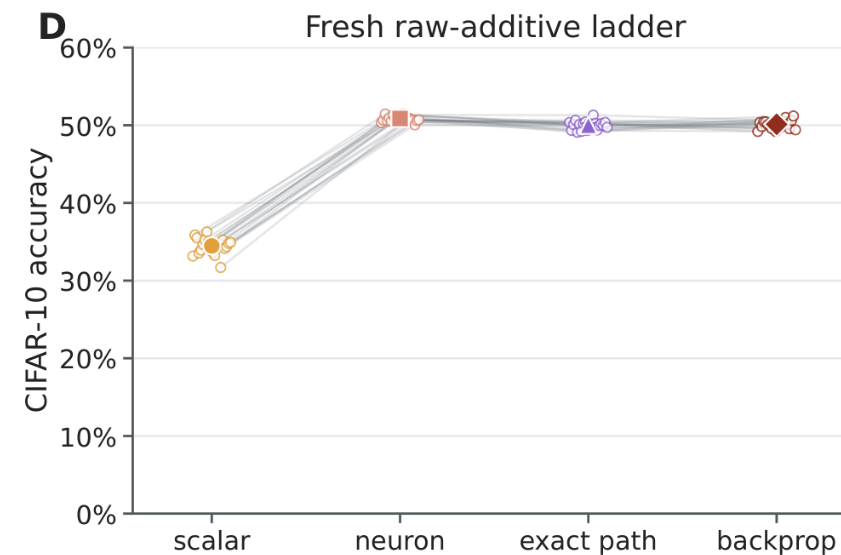
Neuron-specific feedback supplies nearly all useful resolution

MNIST · strict-scalar confirmation

**+0.19 pp**

exact path over neuron-specific feedback on additive MNIST

CIFAR-10 · fresh additive confirmation

**-0.86 pp**

exact path versus neuron-specific feedback on CIFAR-10

no generic gain

harder data do not create a need for within-tree routing

→ On ordinary classification, selecting the correct neuron is the dominant feedback bottleneck.

Restricted feedback helps only when it rejects more noise than signal

$$\mu_M = M(\mu + \xi)$$

OPERATOR UTILITY

$$U(M) = \frac{[\mu^T M \mu]^2}{2L [\|M\mu\|^2 + \text{tr}(M \Sigma M^T)]}$$

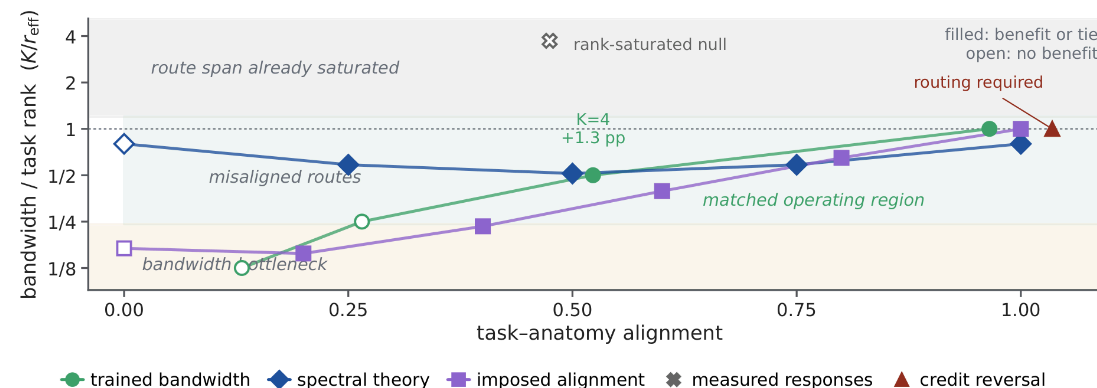
retained signal

finite-step cost

admitted noise

$$\rho = 0.937$$

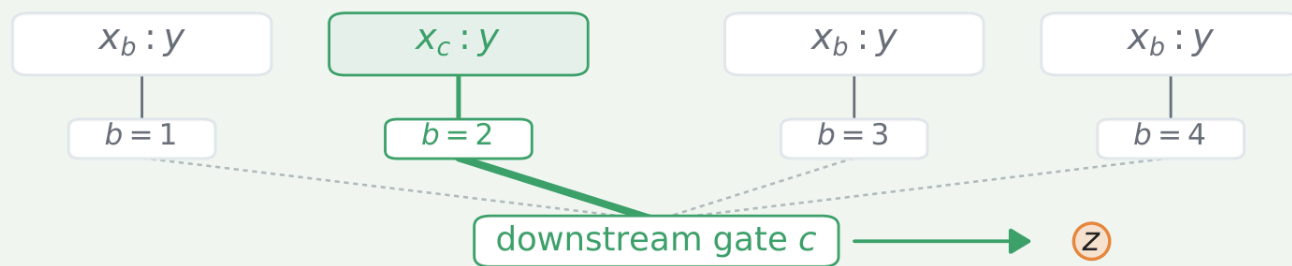
utility versus observed one-step progress across trained conditions



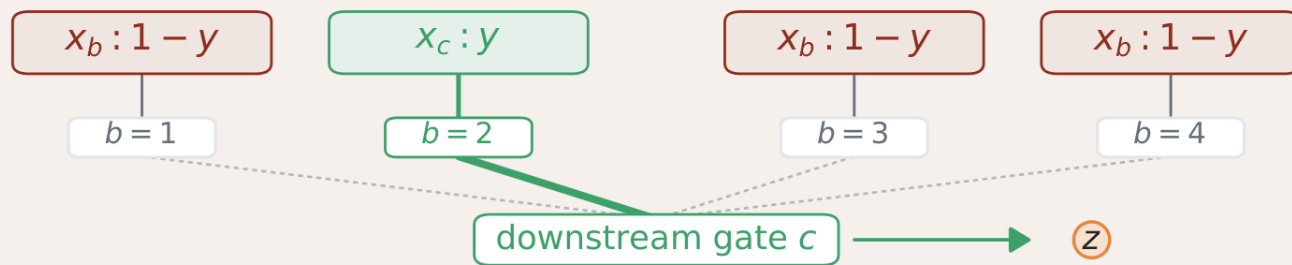
→ Useful routing requires both task alignment and the right bandwidth; more detailed feedback is not always better.

Credit conflict creates a demand for branch-specific signals

compatible, $\chi = 0$



conflicting, $\chi = 1$



input

All B branches receive a Fashion-MNIST image and form nonzero eligibility.

forward selector

Context c selects the branch whose image determines the target.

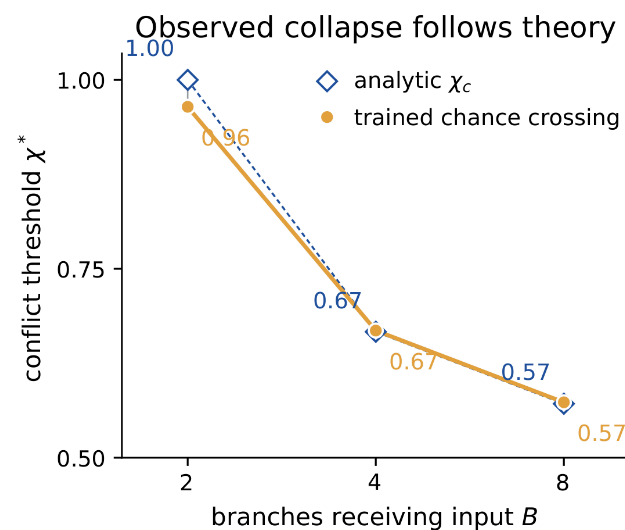
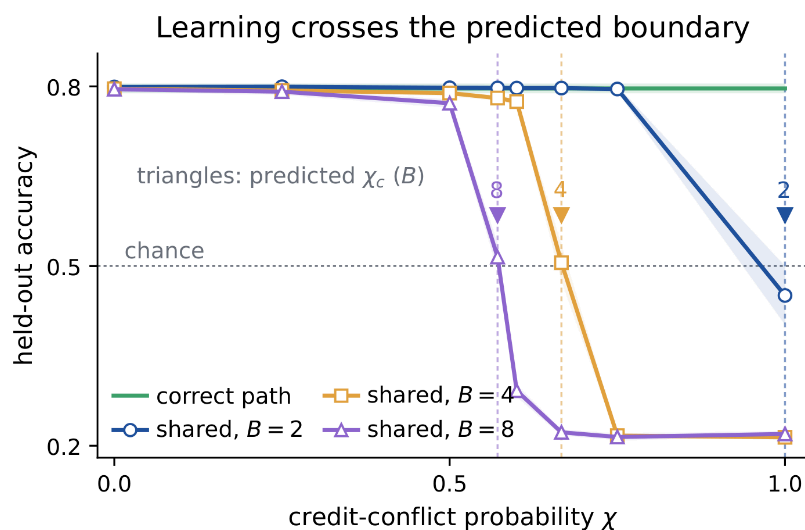
conflict dose χ

Nonselected images vary from class-compatible ($\chi=0$) to opposite-class ($\chi=1$).

$$d_b^{\text{branch}} \propto \sum_i \delta_i [c_i = b] x_{i,b}$$

→ At zero conflict, one shared signal is sufficient; at high conflict, different branches require opposite updates.

Shared credit fails at the predicted conflict boundary



$$\chi_c = B / [2(B-1)]$$

35–58 pp

branch-specific gain over shared feedback at full conflict

causal control

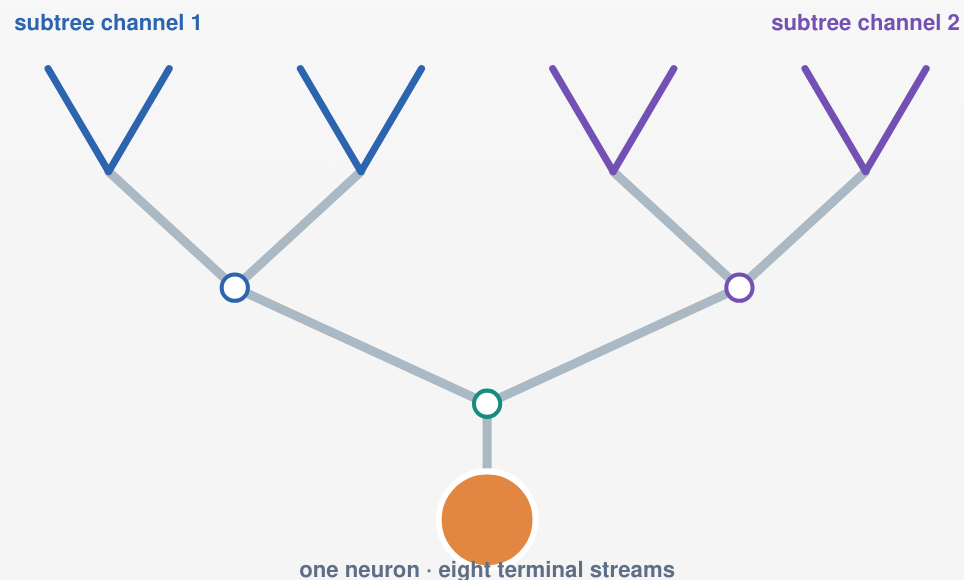
Cyclically deranged routes fail despite identical rank and sparsity.

implementation control

A context-gated point model matches correct routing.

→ The task establishes a need for an address—not a unique need for dendritic material.

Nested tasks ask whether a few subtree addresses are efficient



K = 1 one signal for the whole neuron

K = 2 two coarse subtrees

K = 4 four intermediate subtrees

K = 8 one signal per terminal branch

task hierarchy

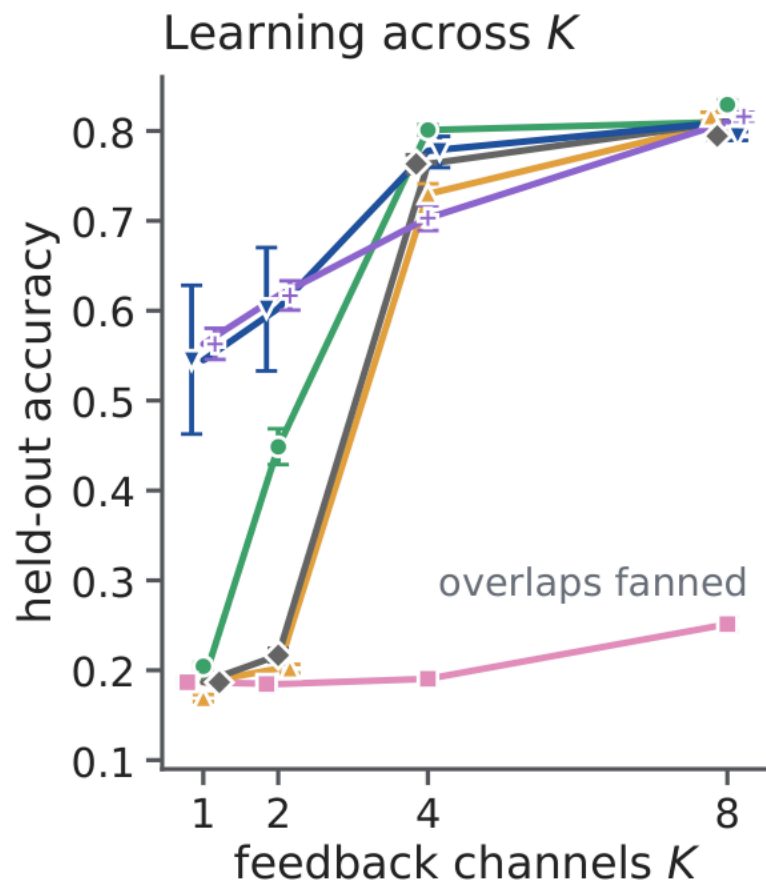
Eight active streams occupy the leaves; opposite-sign distractors strengthen with tree distance.

matched controls

Same rank, sparsity, parameter count, and forward resources; only route assignment or basis changes.

→ Intermediate K tests whether tree-structured feedback is a useful low-dimensional basis for task credit.

Subtree addresses help—but fine topology adds only a narrow gain



+61 pp

matched subtrees over one neuron-shared signal at $K=4$

+1.3 pp

matched subtrees over the strongest non-anatomical low-rank control at $K=4$

boundary

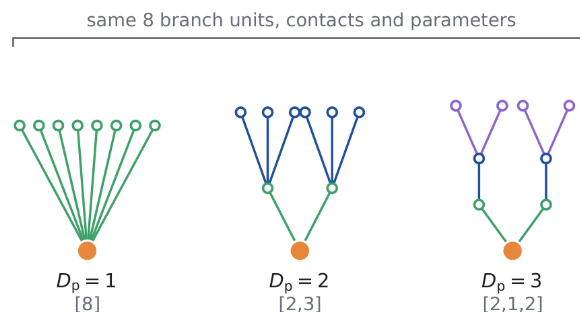
Matched subtrees lose at low K and tie at full rank. Rewiring removes the intermediate advantage.

equivalence

Dendritic, grouped-point, and gated-point implementations coincide for the same routed field.

→ The large effect is the value of within-neuron addresses; the anatomy-specific effect is conditional and modest.

Physical depth helps only when serial computation matches the task



nested factors

flat factors

local ratios

D_p = serial physical stages · H = task hierarchy · α = task-sensor alignment

+30.9 pp

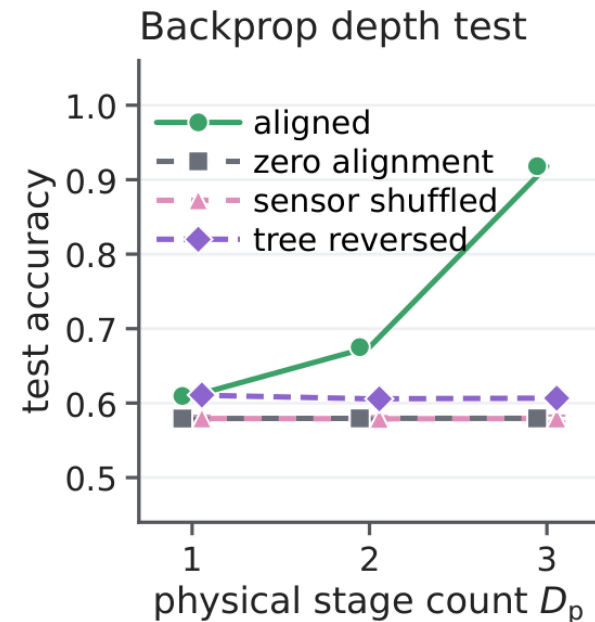
aligned serial BP: D3 over D1 on the calibrated $H=3$ task

+11.0 pp

exact compartment feedback over one shared somatic signal at D3

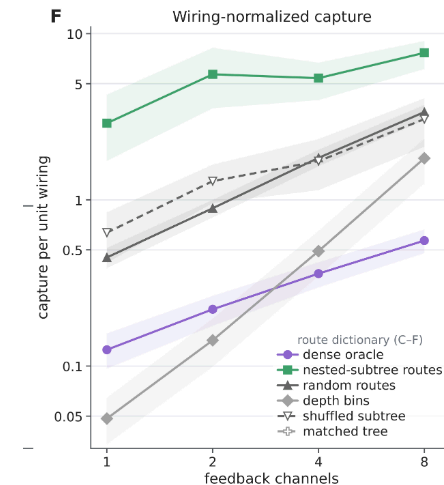
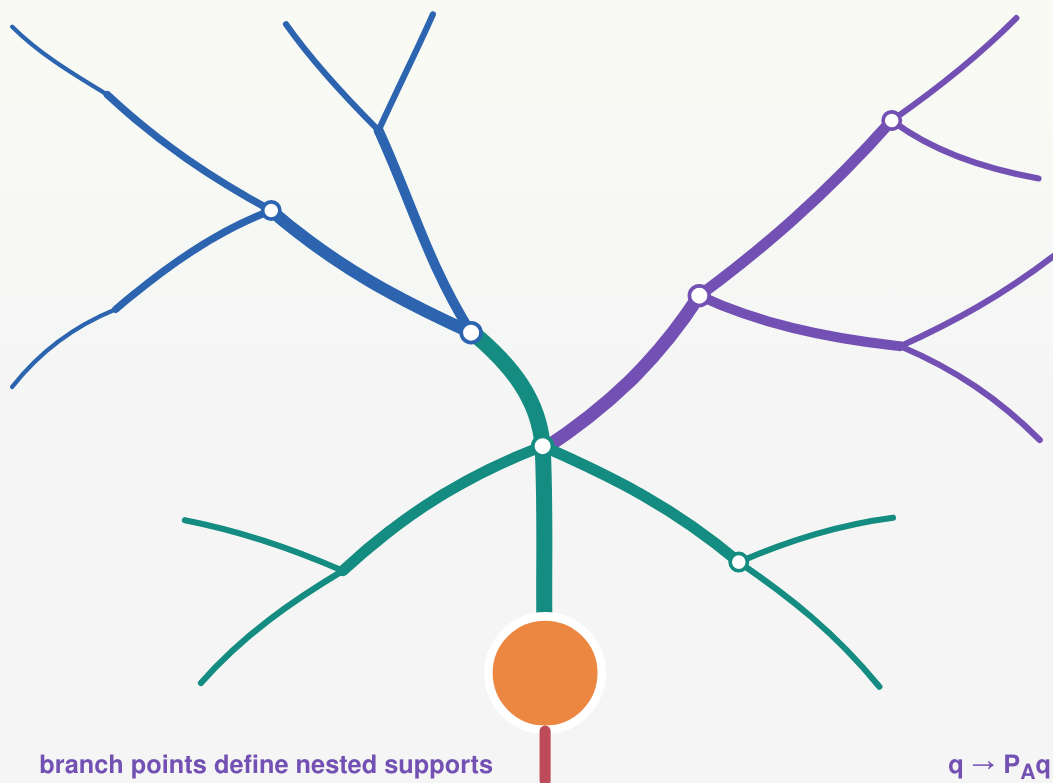
not universal

useful depth saturates and can hurt on local-ratio tasks



➔ Depth is a task-matched inductive bias under constraints, not a general expressivity advantage over point networks.

Real arbors provide sparse candidate routes, mostly through coarse geometry



85%

of dense field capture

7%

of dense feedback connections

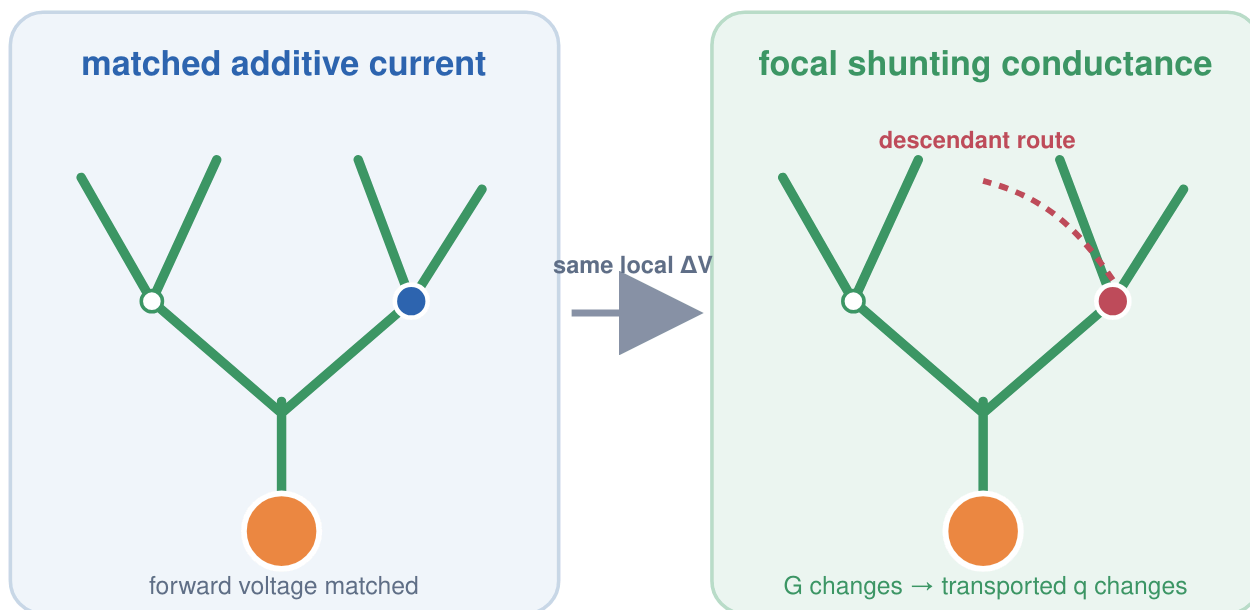
≈ 2.8×

cellwise capture per connection
over density-matched shuffled
routes

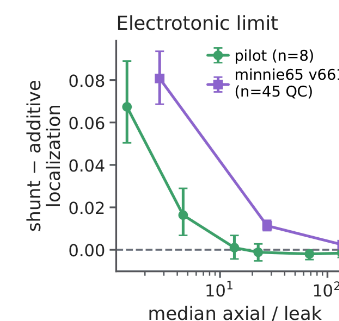
Replicated in a disjoint 47-cell cohort and ten cells from a second MICrONS mouse. These are modeled fields on measured anatomy—not observed task gradients.

➔ Morphology supplies a sparse route dictionary; most capacity comes from branch depth and coarse topology.

Focal shunting changes descendant credit only in permissive regimes



$$q' = q - [\kappa q_k / (1 + \kappa(G^{-1})_{kk})] G^{-1}e_k$$



standard passive calibration

The shunt-minus-current localization contrast is effectively zero.

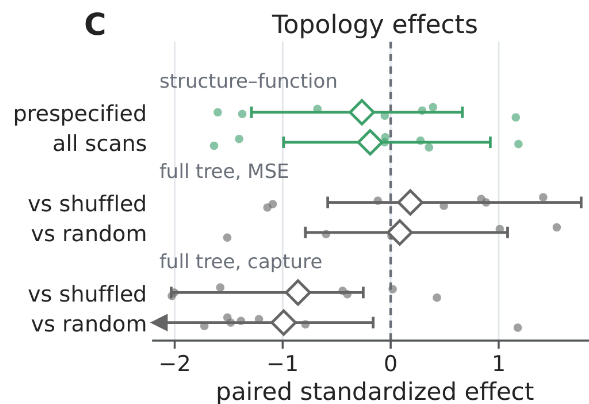
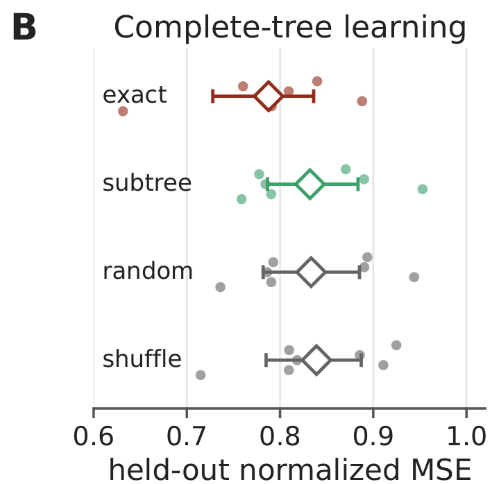
high-conductance regime

Descendant-localized changes emerge and survive active-channel extensions.

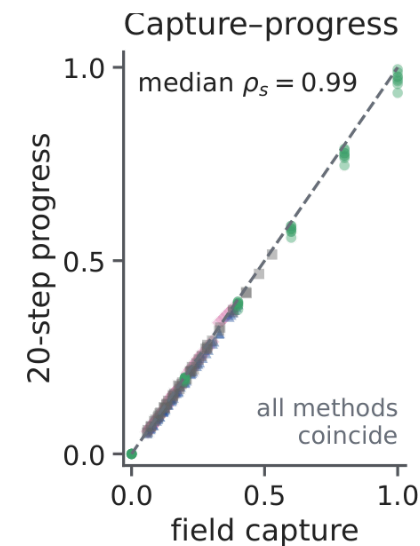
→ Shunting is a state-dependent route-gain mechanism, not a generic learning advantage.

Measured responses are null; imposed task alignment rescues the same routes

Measured MICrONS visual responses



Controlled rotation into the subtree span



what the data show

Nested subtrees do not beat random or site-shuffled routes for held-out response prediction or field capture.

what the rescue shows

Holding anatomy and field energy fixed, capture rises continuously as the task field rotates into the subtree span.

→ The routes are sufficient when aligned, but their endogenous use is not established by the measured visual-response task.

Dendrites provide conditional resources for local credit

neuron identity

ordinary tasks + animal coordinate



subtree addresses

large only under task-aligned conflict



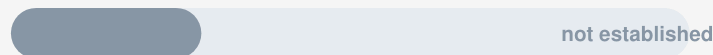
route gain by shunting

requires a high-conductance regime



endogenous morphology alignment

measured-response analyses are null



1 · identify the neuron

One learning coordinate per neuron carries most of the practical benefit on ordinary tasks.

2 · address the subtree

Useful when branches require different task-dependent updates and route bandwidth is matched.

3 · regulate route gain

Conductance can shape transport, but only in an appropriate electrotonic state.

The decisive variable is **alignment between task credit and the available route span**.

→ Dendritic structure is a conditional substrate for routing local credit—not a general replacement for backpropagation.